

Probability and Statistics

Random Experiments and Sample Spaces:-

↓
Experiments involving
Randomness
Coin toss, dice roll, etc.

→ Ω → Set of all possible outcomes
of a random experiment. It could be
finite or infinite.

$$\Omega_c = \{\text{Head, Tail}\}, \Omega_d = \{1, 2, 3, 4, 5, 6\}$$

$$2\text{-coins} \rightarrow \Omega_{2c} = \Omega_c \times \Omega_c = \{H, T\} \times \{H, T\}$$

Outcomes and Events

↓
(omega) $\rightarrow \omega \in \Omega$ is called a sample point
or a possible outcome

↓
A subset $A \subseteq \Omega$ is called an event.

* Events in the coin toss experiment $C_1 = \{T\} [\subseteq \Omega_c]$
Events in the die roll $D_1 = \{6\}, D_2 = \{1, 3, 5\} [\subseteq \Omega_d]$

Events are any subset of Ω , even null sets, but $P(\emptyset) = 0$

Probability of an event $A = P(A)$.

It may or may not be possible to measure / assign P for every subset A .

$$P_c(T) = 0.5$$

Probability measure P is a set function. It acts on sets and measures the probability of such sets.

Set Theory 101:-

A^c = complement of A

$\emptyset$ = denotes empty set. [belongs to every set]

$A \cup B$ = A union B

$A \cap B$ = A intersection B

~~$A \setminus B$~~

$A \setminus B$ = A minus B = $A \cap B^c$

Symmetric

$A \Delta B = (A \setminus B) \cup (B \setminus A)$

ME = Mutually Exclusive

HW $\rightarrow$ ~~Identity Complement~~

$|A|$ = no. of elements in A = cardinality of A.

Inclusion - Exclusion Principle = $|A \cup B| = |A| + |B| - |A \cap B|$

Countable Sets and Uncountable Sets

Monotone Seq $\rightarrow A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$ [Increasing seq]
 $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$ [Decreasing]

$I_n = [0, 1 - \frac{1}{n}]$

$D_n = [0, \frac{1}{n}]$

Cartesian product :- $A \times B = \{(a, b) : a \in A, b \in B\}$

Powerset $[P(A)]$:- Set of all possible subsets of A.

$|P(A)| = 2^{|A|}$ [Only for discrete elements]

$P([0, 1]) = \{(a, b) : a \leq b, a, b \in [0, 1]\}$

$\hookrightarrow$ range, not just 2 elements

Functions:-

Fns are maps from elements in the domain D to the range R

$f: D \rightarrow R$

$f: \mathbb{R} \rightarrow \mathbb{R} \Rightarrow f(x) = x$

Set fns are those fns where who act on sets; D is a collection

of sets.

IP is a set fn.

Axioms of probability:-

1) $P(\emptyset) = 0$, $P(\Omega) = 1$

2) For a set $A \subseteq \Omega$, $P(A) \in [0, 1]$

3) For a disjoint collection of events $A_1, A_2, \dots, A_j$ where $A_i \subseteq \Omega$, then:

$$P\left(\bigcup_{i=1}^j A_i\right) = \sum_{i=1}^j P(A_i) \quad [j \text{ can be } \infty] \rightarrow \text{but must be countable}$$

where $A_i \cap A_j \forall i, j = \emptyset$

In general, the domain of P is $\mathcal{P}(\Omega)$

~~P Probability of impossible event = 0.~~

But

Counter example for $\mathcal{P}(\Omega)$ being satisfactory

1) Pick a number randomly (uniformly) of $\mathbb{R}$ from the real line.

2) $\Omega = \mathbb{R}$, hence $IP(\mathbb{R}) = 1$.

3) Domain = $\mathcal{P}(\mathbb{R})$

4) $P: \mathcal{P}(\mathbb{R}) \rightarrow [0, 1]$

5) P has the property that sets of equal length have equal probability

6) We know that $\mathbb{R} = \bigcup_{n=-\infty}^{\infty} [n, n+1)$ where $[n, n+1) \in \mathcal{P}(\mathbb{R})$

[Countable union since n is an integer]

7) $IP(\mathbb{R}) = 1 = \sum IP([n, n+1)) = 0 / \infty$ depending

8) This is a contradiction! [Happens when picking segments of unit length]

Powerset is thus a bad choice

Not all set fn can be calibrated to measure all possible subset of the sample space.

Our choice of domain must have certain features.

Let domain = $\mathcal{F}$

then $\emptyset, \Omega \in \mathcal{F}$; if $B \in \mathcal{F} \rightarrow B^c \in \mathcal{F}$ also;

$\uparrow$ Sets that meet these criteria are called sigma-algebras

Sigma algebra / Event space $[\mathcal{F}]$ associated with a set Ω is a collection of subsets of Ω that satisfy

$\rightarrow \emptyset \in \mathcal{F}$ and $\Omega \in \mathcal{F}$

$\rightarrow A \in \mathcal{F} \rightarrow A^c \in \mathcal{F}$

$\rightarrow A_1, A_2, \dots, A_n \in \mathcal{F} \rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

$\rightarrow$ The σ -algebra is closed under formation of complements and countable unions and countable intersections (by De Morgan's)

$\rightarrow$ When Ω is countable and finite, then $\mathcal{P}(\Omega)$ is a σ -algebra

If Ω is countable & finite, we will consider $\mathcal{P}(\Omega)$ as the domain.

Probability Space = $\{\Omega, \mathcal{F}, P\}$

Probability measure:- A probability measure P on the measurable space $(\Omega, \mathcal{F})$ is a set fn $P: \mathcal{F} \rightarrow [0, 1]$ s.t.

1) $P(\emptyset) = 0, P(\Omega) = 1$

2) For a disjoint collection of events $A_1, A_2, \dots$ from $\mathcal{F}$, we have

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

P space for $U[0, 1]$

$\Omega = [0, 1]$

If we generate a σ -algebra, but we get a borel-sigma algebra $\mathcal{B}[0, 1]$

Borel σ -algebra $\mathcal{B}[0,1]$

Defined when $\Omega = [0,1]$, as the σ -algebra generated by closed sets of the form $[a,b]$ where $a \leq b$ & $a, b \in [0,1]$

$$(a,b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}] \quad \text{--- } \left\{ \left[\overset{a+c}{a} \overset{b-c}{b} \right] \right\}$$

$$(a,b] = \bigcup_{n=1}^{\infty} (a, b + \frac{1}{n}]$$

 $\mathcal{B}[0,1]$

Borel σ -algebra $\mathcal{B}$: is the σ -algebra generated by sets of the form $[a,b]$ or (a,b) or $(a,b]$ or even $[a,b)$ where $a \leq b$ & $a, b \in [0,1]$

If $\Omega = \mathbb{R}$, the $\mathcal{B}(\mathbb{R})$ is the σ -algebra generated by open sets of the form (a,b) where $a < b$ and $a, b \in \mathbb{R}$.

How to define $\mathcal{B}(\mathbb{R}^2)$

Consequences of the Probability Axioms:-

i) $P(A^c) = 1 - P(A)$

$$P(A \cup A^c) = P(A) + P(A^c) = 1$$

ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

~~$P(A \cup B)$~~ = Prove it

iii) If $A \subseteq B$, prove that $P(A) \leq P(B)$ ($A \subseteq B$ can be interpreted as $A \rightarrow B$)

$$\text{If } A \subseteq B \rightarrow B = A + \lambda$$

iv) $P\left(\bigcup_{i=1}^{\infty} B_i\right) \leq \sum_{i=1}^{\infty} P(B_i)$ [Boole's ineq.]

Diff b/w Impossible event vs Zero probability event

$\emptyset$ $\leftarrow$

$\downarrow$
finite sized set

In $U[0,1]$, $P(\omega = 0.5) = 0$

Every experiment outcome of this experiment is a zero probability event. This implies that zero outcome events could occur.

$\emptyset$ on the other hand, can never occur, and are hence impossible events.

$$P(\omega \in [0, 0.25] \cap [0.75, 1]) = 0$$

Limits and Continuity:-

Limits:-

Let $a_1, a_2, \dots, a_n$ be a sequence with limit L .

Then $\forall \epsilon, \exists N_\epsilon$ s.t. $\forall n > N_\epsilon, |a_n - L| \leq \epsilon$

For a fn $f(x)$, $\lim_{x \rightarrow c} f(x) = f(c)$

The limit exists only if $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x)$.

All points δ close to c are ϵ close to $f(c)$.

Continuity:-

LHL = RHL = $f(c)$. If the fn is cont at all pts, it is said to be continuous. As $x \rightarrow c$, $f(x) \rightarrow f(c)$

But what about IP?

For a continuous set fn S as $A_n \rightarrow A$, we have $S(A_n) \rightarrow S(A)$

Sequence of sets:-

Given $(\Omega, \mathcal{F})$, if $A_1, A_2, A_3, \dots$ is an increasing sequence of events defined on $\mathcal{F}$ and $\bigcup_{n=1}^{\infty} A_n = A \in \mathcal{F}$, then we say that the given seq. of sets A_n are increasing to A ($A_n \uparrow A$)

Similarly when $A_1 \supset A_2 \supset \dots$ is a decreasing seq. of events and $\bigcap_{n=1}^{\infty} A_n = A$, then we have $A_n \downarrow A$

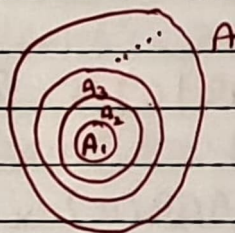
Alt notation:- increasing seq. of sets $A_n \Rightarrow \lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n$

decreasing seq. of sets $A_n \Rightarrow \lim_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} A_n$

Lemma:- For a sequence of events of the type $A_n \uparrow A$ or $A_n \downarrow A$, we have $\lim_{n \rightarrow \infty} P(A_n) = P(A)$

Proof:-

$$A = \bigcup_{i=1}^{\infty} A_i$$



(i) Consider an increasing seq.

$$\begin{aligned} \text{define } F_n &= A_n - A_{n-1} \\ \bigcup_{n=1}^{\infty} A_n &= \bigcup_{n=1}^{\infty} F_n = \sum_{n=1}^{\infty} F_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(F_i) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n F_i\right) = \lim_{n \rightarrow \infty} P(A_n) \end{aligned}$$

Hence Proved.

(ii) Decreasing seq. [H.W.]

Conditional Probability :-

Ex:- Given the die outcome is odd, what is $P(1)$? [$1/3$]
Given $\bar{\omega} \in [0, 0.5]$, what is $P(\bar{\omega} \in [0, 0.25])$

Defn:- The conditional probability of event A is defined as

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \text{ when } P(A) > 0$$

Theorem:- $P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$

$$\frac{P(A \cap B)}{P(B)} \times P(B) = \frac{P(B \cap A)}{P(A)} \times P(A)$$

Bayes' Rule:- $P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$

(i) What is $P(A|B \cap C)$

(ii) $P(A \cap B) = P(A|B) \times P(B)$

(iii) $P(A \cap B \cap C) = P(A|B \cap C) \times P(B \cap C)$

$$\downarrow$$

$$\frac{P(A \cap B \cap C)}{P(C)} \times \frac{P(C)}{P(B \cap C)} \times \frac{P(B \cap C)}{P(C)}$$

$$\downarrow$$

$$P(A|B \cap C) \times P(B \cap C)$$

(iv) $P(A \cap B \cap C) = P(A) \times P(B|A) \times P(C|A \cap B)$

$$\downarrow \qquad \qquad \qquad \uparrow$$

$$\frac{P(A \cap B \cap C)}{P(A \cap B)} \times \frac{P(A \cap B)}{P(A)} \times P(A)$$

iv) $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \times \dots \times P(A_n | A_1 \cap \dots \cap A_{n-1})$

Q) Draw 4 cards without replacement. What is the $P(qc, ed, Ksp, Kcl)$

$$P(qc) \times P(ed|qc) \times P(Ksp|qc \cap ed) \times P(Kcl|qc \cap ed \cap Ksp)$$

$$= \frac{1}{52} \times \frac{1}{51} \times \frac{1}{50} \times \frac{1}{49} = \frac{1}{52!}$$

When every outcome is equally likely in a finite sample space Ω , $P(A) = \frac{|A|}{|\Omega|}$

Law of Total Probability:-

$$A = (A \cap B) \cup (A \cap B^c), \quad P(A) = P(A \cap B) + P(A \cap B^c)$$

$$\text{Same as } P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

Let $B_1, B_2, \dots, B_n$ be the partition of the sample space Ω . Then for any event A we have

$$P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A|B_i)P(B_i)$$

Q) 3 bags with M marbles total. Bag i has R_i red and B_i blue for $i \in [1, 3]$. Find $P(\text{marble} = \text{red from rand bag})$

$$\frac{1}{3} \times \sum_{i=1}^3 \frac{R_i}{R_i + B_i} = \frac{1}{3} \times \sum_{i=1}^3 P(\text{Red} | B_i) \times P(B_i)$$

Independence:-

An event A is independent (w.r.t on event B) if

$$P(A|B) = P(A)$$

Ex:- Simultaneously Tossing a coin and rolling a die

$$\Omega_1 = \{H, T\} \quad \Omega_2 = \{1, 2, 3, 4, 5, 6\}$$

$$\Omega = \Omega_1 \times \Omega_2$$

$$\mathcal{F} = \text{Powerset}(\Omega)$$

$$P(\{H, 6\}) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

$$P(\{T, \text{odd}\}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P(A \cap B) = P(A) P(B) \iff \text{Independence}$$

If A and B are independent

$$P(A^c) = 1 - P(A) \quad ; \quad P(B^c) = 1 - P(B)$$

$$P(A^c) \times P(B^c) = 1 - P(A) - P(B) + P(A \cap B) = 1 - P(A \cup B)$$

Where to now?

$$= P(A \cup B)^c$$

$$= P(A^c \cap B^c)$$

A^c & B^c are independent

$$\text{Prove } P\left(\bigcup_{i=1}^n A_i\right) = 1 - \prod_{i=1}^n (1 - P(A_i))$$

Mutual and Pairwise independence

A collection of events $\{A_i, i \in I\}$ are said to be mutually independent if the $P(\bigcap_{j \in J} A_j) = \prod_{j \in J} P(A_j)$ for any subset $J \subseteq I$

A collection of events $\{A_i, i \in I\}$ are said to be pairwise independent if any pair of events from the collection are independent $\Rightarrow$ all pairs.

Ex:- Pick a random number from $\{1, \dots, 10\}$

$$P(A) = \{\omega \mid \omega < 7; \omega \in \{1, \dots, 10\}\} = \frac{3}{10}$$

$$P(B) = \{\omega \mid \omega < 8; \omega \in \{1, \dots, 10\}\} = \frac{7}{10}$$

$$P(C) = \{\omega \mid \omega \% 2 = 0; \omega \in \{1, \dots, 10\}\} = \frac{1}{2}$$

$$P(A \cap B) = P(A) \rightarrow \text{not independent}$$

$$P(A \cap C) = P(A) \times P(C) \rightarrow \text{independent}$$

$$P(B \cap C) = \frac{3}{10} \neq P(B) \times P(C) \rightarrow \text{not independent}$$

Correlation b/w events:-

Two events A & B are +vely correlated iff $P(A|B) > P(A)$

Two events A & B are -vely correlated iff $P(A|B) < P(A)$

A & B have same correlation as A^c & B^c (Prove)

A & B have opp correlation as A & B^c (Prove)

Mutual Exclusivity:-

Two events are mutually exclusive if $P(A \cap B) = 0$
The occurrence of one implies the other cannot occur.

Two events with non zero probabilities cannot be both mutually exclusive and independent.

$$P(A \cap B) = 0 \neq P(A)P(B) \rightarrow \text{Contradiction.}$$

If A & B are ME

$$P(A|B^c) = \frac{P(A \cap B^c)}{P(B^c)} = \frac{P(A)}{P(B^c)}$$

When $A \subseteq B$, A & B are not ME nor I

Zero Probability Events are always independent

Let $P(E) = 0$

Then for any set F, we want to show that $P(E \cap F) = 0$

But $E \cap F$ has two choices

(a) $E \cap F = \emptyset$

(b) $E \cap F \subseteq E$

In either case $P(E \cap F) = 0$ or $P(E \cap F) \leq P(E) = 0$
 $P(E \cap F) = 0 \leftarrow$

Conditional Independence:-

Two events A & B are conditionally independent of C with $P(C) > 0$ if $P((A \cap B) | C) = P(A | C) \cdot P(B | C)$

It thus follows that $P(A | B \cap C) = P(A | C)$

Ex:- Given a fair coin and a fake coin (2 heads). Pick a random coin and toss twice.

A:- First toss $\rightarrow H = 3/4$

B:- Second toss $\rightarrow H = \text{still } 3/4$

~~Event~~ C :- First coin is chosen = $1/2$

$$P(A \cap B) = P(A \cap B | C) + P(A \cap B | C^c)$$

= $\downarrow$ Fair coin

$\downarrow$ Unfair coin

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$\frac{1}{2}$$

$$= \frac{5}{8} \neq \frac{3}{4} \times \frac{3}{4} \rightarrow \text{Not independent}$$

C: coin is chosen and tossed twice; not one coin being chosen and tossed twice

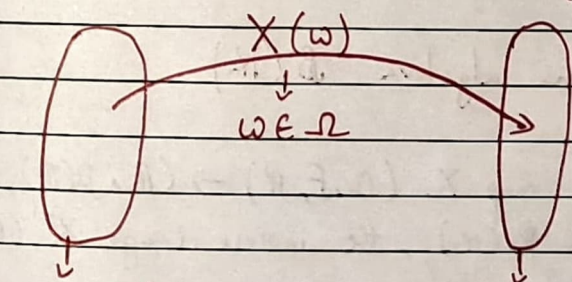
A & B are conditionally independent on C, not C^c

Random Variables

Given a random experiment with associated $(\Omega, \mathcal{F}, P)$, it is sometimes difficult to deal directly with $\omega \in \Omega$. ω is not a number, it is a sequence (vector) of nos.

Random variables are devices that help us make a mapping from $(\Omega, \mathcal{F}, P)$ to $(\Omega', \mathcal{F}', P_x)$, as this would help analyze functions of sample pts rather than any ^{single} sample pt.

Assume you want to count no. of 6s in 10 rolls



$\Omega =$ set of all 10 vectors

$\Omega' = \{1, \dots, 10\}^6$; $\mathcal{F}'$ is the Powerset (Ω')

P

$\otimes P_x$

A random variable X is a fn $X: \Omega \rightarrow \Omega'$ that transforms the probability space $(\Omega, \mathcal{F}, P)$ to $(\Omega', \mathcal{F}', P_X)$ and is $(\mathcal{F}, \mathcal{F}')$ -measurable

The map $X: \Omega \rightarrow \Omega'$ implies $X(\omega) \in \Omega' \quad \forall \omega \in \Omega$

For the event $B \in \mathcal{F}'$, the preimage $X^{-1}(B)$ is defined as $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$

→ For every $B \in \mathcal{F}'$, we have $X^{-1}(B) \in \mathcal{F}$

X^{-1} is a one to many mapping from $\mathcal{F}' \rightarrow \mathcal{F}$

$P_X(B) =$ Probability that X results in $\{B\}$ while acting being acted upon by $\omega \in \Omega$

$=$ Probability of $P(\omega)$ where $\omega \in \Omega; X(\omega) \in B$

$\mathcal{F}$ could have elements that have no ~~B~~ images in $\mathcal{F}'$. But $\forall f \in \mathcal{F}'$, f has a pre image in $\mathcal{F}$

P_X is called induced probability measure

Conventions:-

i) $\Omega' = \mathbb{R}$

ii) $\mathcal{F}' =$ Borel sigma algebra $\mathcal{B}(\mathbb{R})$

A rand var X is a map $X: (\Omega, \mathcal{F}, P) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$ such that for each $B \in \mathcal{B}(\mathbb{R})$, the inverse image $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$ satisfies

$$X^{-1}(B) \in \mathcal{F}$$

$$P_X(B) = P(\omega \in \Omega : X(\omega) \in B) = P(X^{-1}(B))$$

If Ω' is countable, then the random variable is called a discrete random variable. Only here is $\mathcal{F}'$ used as the σ -algebra.

Capital letters XYZ etc for rand var, and x, y, z etc are the realizations of said rand var.

Discrete Random Variables:-

Ex:- Roll two dice and calculate the sum

Ω = set of all tuples (x, y) where $x, y \in [1, 6]$

Ω' = integers $[2, 12]$

$\mathcal{F}$ & $\mathcal{F}'$ are σ -algebras

$$\{X=3\} \in \mathcal{F}' \rightarrow P_X(3) = P(\{1,2\}, \{2,1\})$$

In general for $x \in \Omega'$ $P_X(x) = P(\{a, b\} \in \Omega, a+b=x)$

$$p_X(x) = P_X(\{x\})$$

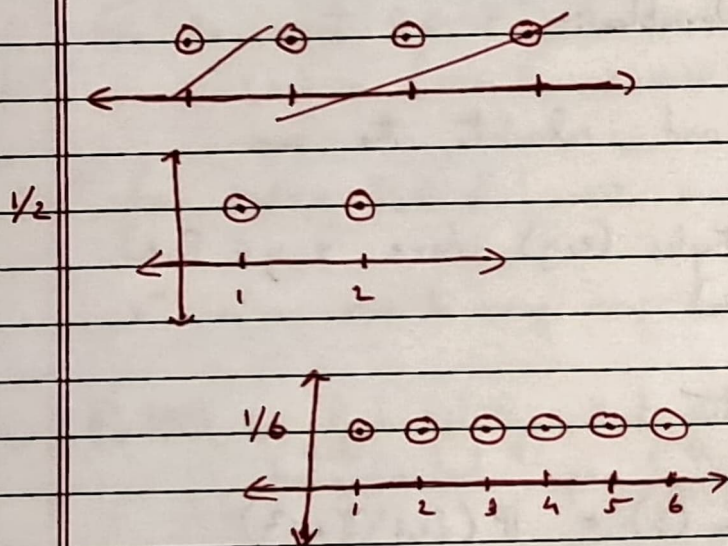
$\rightarrow$ fn [not a probability measure] $\rightarrow$ makes x a set and passes as arg to P_X .
 $\rightarrow$ Probability mass fn.

$$p_X(x) = \begin{cases} x-1/36 & \forall x \in \text{int}[2, 7] \\ 13-x/36 & \forall x \in \text{int}[8, 12] \end{cases}$$

Multiple X_i s can have the same Probability Mass fn.
 For example, for the exp of rolling a die four times, then divide and conquer using $p_X(x)$ mentioned above.
 Only if X_i s are independent

The fn $p(x) := P_X(\{x\})$ for $x \in \Omega'$ is called as a Probability Mass Function (PMF) of a random variable X .

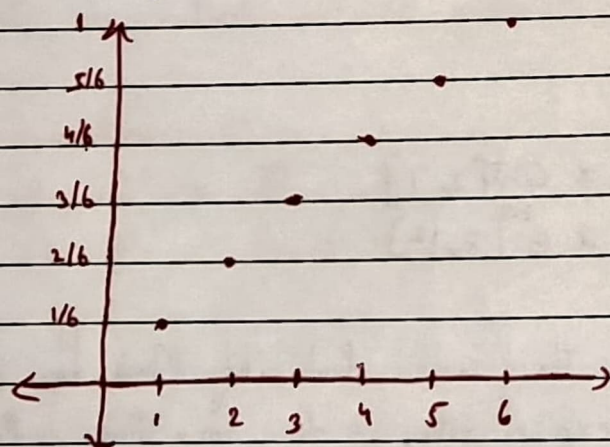
Q) Plot PMFs for X corresponding to coin tosses or dice rolls.



The Cumulative Distribution Function (CDF) is defined as

$$F_X(x_1) = \sum_{x \leq x_1} p(x) = P\{\omega \in \Omega : X(\omega) \leq x_1\}$$

F_X (Rolling dice)



Ω'

The X-axis is only X not Ω . Here the mapping X is identical to Ω .

Expectation and Moments:-

The mean or expectation of a random variable X is denoted by $E(X)$ and is given by $E(X) = \sum_{x \in \Omega'} x p_x(x)$

The expected value need not be part of Ω

Depends on how X is defined.

First moment of a rand var.

The n^{th} moment of a random variable X is denoted by $E(X^n)$ and is given by $E(X^n) = \sum_{x \in \Omega'} x^n p_x(x)$

For a fn $g(\cdot)$ of a rand var X , $E(g(X)) = \sum_{x \in \Omega'} g(x) p_x(x)$

fns of random variables are random variables with their own probability mass fns.

Consistency of PMF:-

PMF: $p_x(x) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\})$ for $x \in \Omega'$

To prove $\sum_{x \in \Omega'} p_x(x) = 1$

$$\begin{aligned} \sum_{x \in \Omega'} p_x(x) &= \sum_{x \in \Omega'} \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}) \\ &= \mathbb{P}\left(\bigcup_{x \in \Omega'} \{\omega \in \Omega : X(\omega) = x\}\right) \\ &= \mathbb{P}(\Omega) = 1 \end{aligned}$$

Expectance is Linear :-

$$\text{Let } Y = aX + b$$

$$\begin{aligned} \text{Then } E(Y) &= E(aX + b) \\ &= \sum_{x \in \Omega'} (ax + b) p_x(x) \\ &= a \left(\sum_{x \in \Omega'} x p_x(x) \right) + b \\ &= a E(X) + b \end{aligned}$$

The f_n is OTO, then you could still apply this }
 $y_i = ax_i + b$ }

$$P(y_i) = P(x_i) \rightarrow PMF(y_i) = PMF(x_i)$$

$$\begin{aligned} E[Y] &= \sum_{y \in \Omega'} y p_y(y) \\ &= \sum_{x \in \Omega'} (ax_i + b) p_y(ax_i + b) \\ &= \sum_{x \in \Omega'} (ax_i + b) p_x(x) = \sum_{x \in \Omega'} (ax_i + b) p_x(x) \\ &= a E[X] + b \end{aligned}$$

What ~~$y = g(x)$~~ ^{is} $Y = g(x)$ is MTO?

$$\text{Ex:- } Y = |X|$$

Function of Random Variables :-

$$\begin{aligned} \text{Consider } Y &= |X|, X = (\text{int}) [-4, 4] \\ p_x &= \frac{1}{9} \quad \forall x \in X = (\text{int}) [-4, 4] \end{aligned}$$

$$p_y(2) = \sum_{x: |x|=2} p_x(x) = p_x(-2) + p_x(2) = \frac{2}{9}$$

Prove:- If $Y = g(X)$ and PMF of $X = p_x(x)$, then

$$p_y(y) = \sum_{\{x: g(x)=y\}} p_x(x)$$

Theorem:- Suppose $Y = g(X)$ and X is discrete with PMF $p_x(x)$. Then $E[Y] = \sum_x g(x) p_x(x)$

Proof:- $E[Y] = \sum_y y p_y(y)$

$$= \sum_y \sum_{x: g(x)=y} g(x) p_x(x)$$

↓ This is nothing but the summation over all x in domain

$$= \sum_x g(x) p_x(x)$$

Variance:-

Consider $g(x) = (x - E[X])^2$

$g(x)$ quantifies the deviation of x from the mean.

This is called the variance.

$\sqrt{E(g(x))}$ quantifies the deviation

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - E[X]^2$$

↓

$$E[X^2] + E[X^2 - 2XE[X] + E[X]^2] = E[X^2] + E[-2XE[X]] + E[E[X]^2]$$

$$= E[X^2] - 2E[X]^2 + E[X]^2 = E[X^2] - E[X]^2$$

$\sigma_x = \sqrt{\text{Var}(X)}$ is called the Standard Deviation of X

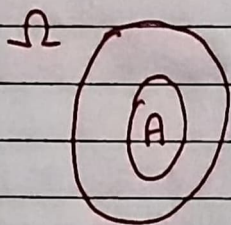
When $Y = aX + b$, $\text{Var}(Y) = a^2 [\text{Var}(X)]$

a^2 cause variance squares
substitute and verify.

a cause variance is not affected by const.

Examples of DRV

(i) Indicator Random Variables

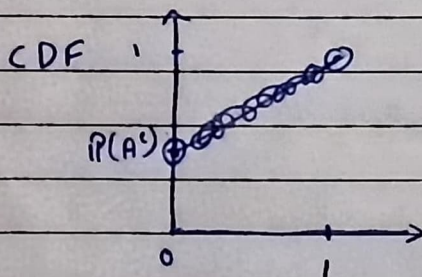


We want to know if A has occurred.

$$I_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A \\ 0, & \text{else} \end{cases}$$

$$P_{I_A}(1) = P(A)$$

$$P_{I_A}(0) = P(A^c)$$



$$\text{Mean} = P(A) (?)$$

(ii) Bernoulli Random Variable

$$X = \begin{cases} 1, & \text{with IP} = p \\ 0, & \text{else} \end{cases}$$

Used Bernoulli Bandits

$$\text{Mean} = p, \quad E[X^n] = p$$

(iii) Binomial Random Variables $B(n, p)$

Treat as a Bernoulli Random Variable

Consider a biased coin ($P(\text{head}) = p$) and toss it n times.

Denote head = 1, tail = 0

Let N denote no. of heads in n tosses

$$p_N(k) = \underbrace{{}^n C_k}_{\substack{\text{no. of ways to} \\ \text{choose } k \text{ coins}}} \cdot \underbrace{p^k}_{\substack{\text{all heads}}} \cdot \underbrace{(1-p)^{n-k}}_{\substack{\text{remaining } n-k \text{ tails}}}$$

$$\text{Mean} = \sum_k k p \quad E[N], E[N^2], \text{Var}(X)$$

(iv) Geometric Random Variable

Consider a biased coin ($P(H) = p$) and keep tossing till you achieve the first head.

Random Variable N = no. of tosses reqd for 1st head

$$p_N(k) = (1-p)^{k-1} \times p$$

$$\bar{F}_N(k) = 1 - F_N(k) = P(N > k)$$

$$P(N > k) = k \text{ tails} = (1-p)^k$$

If head doesn't occur in k tosses, the probability that we need m more tosses ~~of one~~ is independent of k . This is the memoryless property of Geometric Random Variables.

Continuous Random Variables

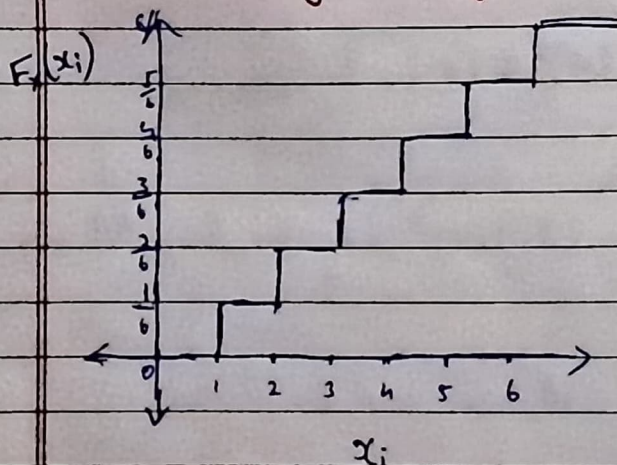
Consider a random variable X that maps $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$ where $\mathcal{B}(\mathbb{R}) =$ Borel σ algebra of $\mathbb{R}$

We then define the CDF by

$$F_X(x) = P\{\omega \in \Omega : X(\omega) \leq x\}$$

Take the set $(-\infty, x_i]$. $P_X((-\infty, x_i]) = F_X(x_i)$

This holds for discrete random variables as well, but the probability of range is the $\sum P(x) \forall x \in \text{range}$.



CDF for die roll assuming the above Rand Variable

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If the CDF graph has discontinuities, the random variable is discrete and vice versa.

CDFs must be monotonically increasing (or staying const) and it must hit 1 at some point.

Mixed Random Variables have cont. & disc. parts in the CDF graph.

All CDFs for continuous rand variables (we consider) is differentiable.

Derivative of the CDF = Density fn (How much is conc. over a len)

Ex of CRV $\rightarrow$ Pick a no. from $[a, b]$

$\hookrightarrow$ level of water in a dam

$\hookrightarrow$ Server work load.

Probability Density fn:- (unit $\rightarrow$ probability per unit len)

$$\text{PDF} \Rightarrow f_x(x) = \lim_{\Delta \rightarrow 0^+} \frac{P_x(x < X \leq x + \Delta)}{\Delta} \rightarrow \frac{P(A \cap B^c)}{\Delta}$$

Any value works,
we just normalize

$$= \lim_{\Delta \rightarrow 0^+} \frac{F_x(x + \Delta) - F_x(x)}{\Delta} \quad \begin{matrix} \downarrow \\ P(A) - P(B) \end{matrix}$$

$$= \frac{d F_x(x)}{dx} \quad (\text{if derivative exists})$$

$$\text{Equivalently} \quad F_x(x) = \int_{-\infty}^x f_x(u) du$$

Properties:-

$$(i) P_x(\mathbb{R}) = \int_{-\infty}^{\infty} f_x(u) du = 1$$

$$(ii) P_x([a, b]) = F_x(b) - F_x(a) = \int_a^b f_x(u) du$$

$$(iii) \text{ In general, for any } B \subseteq \mathbb{R}, P_x(B) = \int_{u \in B} f_x(u) du$$

$$(iv) P_x(\{a\}) = 0 \quad (\text{no mass at any point})$$

↳ Contradiction proof

$$\text{↳ or } \int_a^a f_x(u) du = 0 \quad \text{by defn}$$

$$(v) P_x([a, b]) = P_x((a, b]) = P_x([a, b)) = P_x((a, b))$$

Mean, Variance & Moments:-

$$E[X] = \int_{-\infty}^{\infty} u f_x(u) du$$

$$E[X^n] = \int_{-\infty}^{\infty} u^n f_x(u) du$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(u) f_x(u) du$$

$$\text{Var}[X] = E[g(X)] \text{ \& } g(X) = (X - E[X])^2$$

For $Y = aX + b$

$$(i) E[Y] = a E[X] + b$$

$$(ii) F_Y[Y] = F_X\left(\frac{y-b}{a}\right) \quad (\text{only } a > 0)$$

$$a < 0 \rightarrow 1 - F_X\left(\frac{y-b}{a}\right)$$

Examples:-

(i) Picking uniformly from $[a, b]$

$$f_x(x) = \frac{1}{b-a} \quad \forall x \in [a, b]$$

$$CDF = \int_{-\infty}^x \frac{1}{b-a} du = \left. \frac{1}{b-a} u \right|_{-\infty}^x = \frac{x-a}{b-a}$$

$$CDF \Rightarrow F_x(x) = \begin{cases} 0 & ; x < a \\ \frac{x-a}{b-a} & ; x \in [a, b] \\ 1 & ; \text{otherwise} \end{cases}$$

$$E[X] = \frac{a+b}{2} \quad \text{Var} = \frac{(b-a)^2}{12}$$

(ii) Exponential Random Variable ~~the~~ ($\text{Exp}(\lambda)$)

$$f_x(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

$$CDF = F_x(x) = 1 - e^{-\lambda x} \quad \forall x \geq 0$$

$$E[X] = \frac{1}{\lambda} \quad \text{Var} = \frac{1}{\lambda^2} \quad E[X^n] = \frac{n!}{\lambda^n}$$

Building blocks of Markov Chains.
Are Memoryless.

$$P(X > a+h | X > a) = \frac{e^{-\lambda(a+h)}}{e^{-\lambda a}} = e^{-\lambda h} = P(X > h)$$

(iii) Gaussian Random Variable

$$N(\mu, \sigma^2)$$

Mean $\downarrow$ Variance $\downarrow$ } Two parameters, chance causes the output f_X

$$f_X(x) = \frac{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}{\sqrt{2\pi}\sigma} \quad \forall x \in \mathbb{R}$$

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$E(X) = \mu \text{ \& Var}(X) = \sigma^2$$

(iv) Standard Normal Random Variable

$$N(0, 1)$$

$$\mu = 0, \sigma = 1 \rightarrow \text{standard normal}$$

$$f_X(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$$

$$\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$$

$$F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt = \Phi(x)$$

Normality is preserved under linear transformations

$$Y = aX + b$$

$$\begin{aligned}
 d(u/v) &= d(uv^{-1}) = u dv^{-1} + v^{-1} du \\
 &= -u v^{-2} v' + v^{-1} du \\
 &= \frac{u'v - uv'}{v^2}
 \end{aligned}$$

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$$X \sim N(\mu, \sigma^2)$$

$$E[Y] = a\mu + b$$

$$\text{Var}[Y] = a^2 \sigma^2$$

It hence follows that $Y \sim N(a\mu + b, a^2 \sigma^2)$

Central Limit Theorem:-

$$\frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ where } X_i \text{ is } \text{any random } \overset{\text{independent}}{\text{variable}}$$

variable with mean = μ & var = σ^2

↳ Doesn't need to belong to the same family either, but we will not deal with that for the sake of simplicity.

We'll deal with CLT later in the course.

Gaussian processes are used in Bayesian Optimization

Functions of CRVs

$$\text{Let } Y = ax + b$$

$$F_Y(Y) = P(Y < y) = P(ax + b \leq y)$$

$$F_Y(Y) = F_X\left(\frac{y-b}{a}\right) \text{ if } a > 0 \text{ else } 1 - F_X\left(\frac{y-b}{a}\right)$$

$$f_Y(Y) = F'_Y(Y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) \text{ when } a > 0$$

$$\text{else } -\frac{1}{a} f_X\left(\frac{y-b}{a}\right) \neq$$

$$\text{In general } f_Y(Y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

Hyderabad? No, Bhubaneswar!

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Monotone & continuous $\rightarrow$ invertible

$$Y = g(X) \quad \{\text{Monotonically increasing}\}$$

$$\downarrow \quad \quad \quad \uparrow f_X$$
$$X = h(Y) \rightarrow f_Y(Y) = \frac{d h(Y)}{d Y} f_X(h(Y))$$

Proof:-

$$F_Y(Y) = P(g(X) \leq y)$$

The two events $g(X) \leq y$ & $X \leq h(Y)$ are the same because of the monotonicity of $g(X)$ [and the continuity of $g(X)$ as well], hence their probabilities are the same.

(i) $g(X)$ is non-decreasing:-

$$F_Y(Y) = P(g(X) \leq y) = P(X \leq h(Y))$$
$$= F_X(h(Y))$$

$$f_Y(Y) = \frac{d}{dY} (F_X(h(Y))) = f_X(h(Y)) \frac{d(h(Y))}{dY}$$

(ii) $g(X)$ is decreasing:-

$$F_Y(Y) = P(g(X) \leq y) = P(X > h(Y))$$
$$= 1 - F_X(h(Y))$$

$$f_Y(Y) = \frac{-d}{dY} (F_X(h(Y))) = -f_X(h(Y)) \frac{d(h(Y))}{dY}$$

What if g is not monotone?

Q) $Y = X^2$, then what is $f_Y(Y)$ in terms of $f_X(X)$?

For a random variable X , the CDF F_X satisfies the following:-

$$F_X(\infty) = 1; F_X(-\infty) = 0 \text{ \& when } P_{-}(-\infty < X < \infty) = 1$$

$F_X: \mathbb{R} \rightarrow [0, 1]$ is non-decreasing & right continuous.

At point x of discontinuity, we have

$$\text{right hand limit } F_X(x+) = \lim_{\epsilon \rightarrow 0^+} F_X(x+\epsilon)$$

$$\text{left hand limit } F_X(x-) = \lim_{\epsilon \rightarrow 0^-} F_X(x-\epsilon)$$

$$F_X(x-) \neq F_X(x+)$$

$F_X(x)$ could be either one of them.

Right continuity mandates that at pt of disc, $F_X(x) = F_X(x+)$

$$\text{If } F_X \text{ is continuous, } F_X(x) = F_X(x+) = F_X(x-)$$

Proof of F_X being non-decreasing.

Consider $a < b$ where a, b are chosen arbitrarily.

$$\text{Define } A := \{\omega \in \Omega : X(\omega) \leq a\}$$

$$B := \{\omega \in \Omega : X(\omega) \leq b\}$$

$$\text{Since } A \subseteq B, \quad P(A) \leq P(B)$$

$$F_X(a) = P_X((-\infty, a]) = P(A) \leq P(B) = P_X((-\infty, b]) = F_X(b)$$

Hence proved that F_X is non decreasing.

Proof of F_X being right continuous.

Consider a sequence of no.s $\{x_n\}$ decreasing to x . In this case, we have $F_X(x+) = \lim_{x_n \rightarrow x^+} F_X(x_n)$

$$\text{Let } A := \{\omega : X(\omega) \leq x\} \text{ \& } A_n := \{\omega : X(\omega) \leq x_n\}$$

Clearly $A_n \downarrow A$

From continuity of probability

$$\lim_{n \rightarrow \infty} P(A_n) = P(A)$$

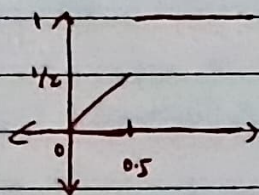
This implies that $\lim_{x_n \rightarrow x} F_X(x_n) = F_X(x)$

You cannot prove the other way by considering $x_n \uparrow x$ because $\bigcup_n (-\infty, x_n] = (-\infty, x) \rightarrow P_X((-\infty, x)) \neq F_X(x)$

Mixed Random Variables:-

Random Variables that are neither continuous nor discrete are called mixed random variables.

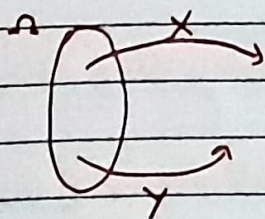
Ex:- X is a random variable $X \sim U[0,1]$ and $Y = X$ if $X \leq 0.5$ & $Y = 0.5$ if $X > 0.5$.



Let $F_Y(y) = c(y) + D(y)$ where $c \rightarrow$ Continuous part,
 $D \rightarrow$ Discontinuous part.

$$E[Y] = \int_{-\infty}^{\infty} x c(x) dx + \sum_{y_i} y_i P(Y = y_i)$$

where $\{y_i\}$ are jump points of $D(y)$ where $P(Y = y_i) > 0$

DiscreteMultiple Random Variables:

Consider the exp of tossing a coin & rolling a die

$$\Omega = \{0, 1\} \times \{1, 2, \dots, 6\}, \quad \mathcal{F} = 2^\Omega, \quad P(\omega) = \frac{1}{12}$$

Let X be the rand var to depict the coin toss & Y depict the die roll

For $\omega = (1, 5)$, we have $X(\omega) = 1$ & $Y(\omega) = 5$

$$X(\omega) = \{0, 1\}$$

$$Y(\omega) = \{1, 2, \dots, 6\}$$

We want to find the joint PMF $p_{xy}(x, y)$ & joint CDF $F_{xy}(x, y)$ of X & Y together.

$$p_{xy}(x, y) := P\{\omega \in \Omega : X(\omega) = x \text{ \& \& } Y(\omega) = y\}$$

$$F_{xy}(x, y) := P\{\omega \in \Omega : X(\omega) \leq x \text{ \& \& } Y(\omega) \leq y\}$$

$$P_*((x, y) \in A) = P\{\omega \in \Omega : (X(\omega), Y(\omega)) \in A\}$$

$$= \sum_{(x, y) \in A} p_{xy}(x, y)$$

Marginals:-

Toss a coin and roll a die-

$$\text{What is } p_{xy}(1,1)? = \frac{1}{12}$$

$$\sum_i p_{xy}(1,i)? = \frac{1}{2} = P\{\omega \in \Omega : X(\omega) = 1\} = \frac{1}{2} = p_x(x)$$

$$\text{Marginalization} \rightarrow p_x(x) = \sum_y p_{xy}(x,y)$$

The marginal PMF, p_x & p_y can be obtained from the joint PMF as follows

$$p_x(x) = \sum_y p_{xy}(x,y) \quad \& \quad p_y(y) = \sum_x p_{xy}(x,y)$$

$$F_{xy} = P(X \leq x, Y \leq y) = P(\omega \in \Omega; X(\omega) \leq x, Y(\omega) \leq y)$$

$$F_{xy}(x_i, y_i) = \sum_{\substack{x \leq x_i \\ y \leq y_i}} p_{xy}(x,y)$$

$$= \sum_{x \leq x_i} \sum_{y \leq y_i} p_{xy}(x,y)$$

Proof:-

$$p_{xy}(x,y) = P(\omega \in \Omega; X(\omega) = x, Y(\omega) = y)$$

$\xrightarrow{\text{marginal}} p_x(x)$

$$\sum_y p_{xy}(x,y) = P\left(\bigcup_y \{\omega \in \Omega; X(\omega) = x, Y(\omega) = y\}\right)$$

$$= p_x(x)$$

Follows that

$$F_x(x) = \sum_y F_{xy}(x,y)$$

Summing over all y of p_{xy} is the same as p_x itself

Independence of Random Variables

If $p_x(x) p_y(y) = p_{xy}(x, y)$ and

$F_x(x) F_y(y) = F_{xy}(x, y)$, we say X and Y are independent
 $E[XY] = E[X] E[Y]$

$$p_{xy}(x, y) = P(\omega \in \Omega; X(\omega) = x, Y(\omega) = y)$$

$$A = \{\omega \in \Omega; X(\omega) = x\}$$

$$B = \{\omega \in \Omega; Y(\omega) = y\}$$

$$P(A) = p_x(x), \quad P(B) = p_y(y)$$

$$p_{xy}(x, y) = P(A \cap B) = P(A) \times P(B) \\ = p_x(x) p_y(y)$$

$$F_{xy}(x, y) = P(X \leq x, Y \leq y)$$

$$= P(\omega \in \Omega; X(\omega) \leq x, Y(\omega) \leq y)$$

A & B would then be $\{\omega \in \Omega; X(\omega) \leq x\}$ & $\{\omega \in \Omega; Y(\omega) \leq y\}$

• Proof proceeds in a similar fashion.

$$E[XY]$$

$$E[X] = \sum_x x p_x(x) = \sum_x \sum_y x p_{xy}(x, y)$$

$$E[Y] = \sum_y y p_y(y) = \sum_y \sum_x y p_{xy}(x, y)$$

$$E[XY] = \sum_x \sum_y xy p_{xy}(x, y)$$

When X, Y are independent, $E[XY] = E[X] \times E[Y]$

Q) Consider rolling a die

$$X = \begin{cases} 1, & \text{outcome is odd} \\ 0, & \text{else} \end{cases}$$

$$Y = \begin{cases} 1, & \text{outcome is even} \\ 0, & \text{else} \end{cases}$$

What is are $p_x(x)$, $p_y(y)$, $p_{xy}(x,y)$ & $F_{xy}(x,y)$

$$p_x(x) = \begin{cases} 1, & 1/2 \\ 0, & 1/2 \end{cases}$$

$$p_y(y) = \begin{cases} 1, & 1/2 \\ 0, & 1/2 \end{cases}$$

$$p_{xy}(x,y) = \begin{cases} 0,0 & ; 0 \\ 0,1 & ; 1/2 \\ 1,0 & ; 1/2 \\ 1,1 & ; 0 \end{cases}$$

Visibly dependent

$$E[XY] = \sum_{x,y} xy p_{xy}(x,y)$$

$$= 0$$

$$E[X] = 1/2$$

$$E[Y] = 1/2$$

Consistency Conditions

$$(i) \sum_{x,y} p_{xy}(x,y) = 1$$

$$(ii) F_{xy}(\infty, \infty) = 1$$

$$(iii) F_{xy}(-\infty, \infty) = F_{xy}(\infty, -\infty) = F_{xy}(-\infty, -\infty) = 0$$

$$(iv) F_{xy}(x, \infty) = F_x(x)$$

$$(v) F_{xy}(\infty, y) = F_y(y)$$

→ Marginal CDFs

Multiple Continuous Random Variables

Pick a random point on the unit square centered at $(0.5, 0.5)$

Random variables X, Y represent the respective x & y coordinate of the point chosen.

$F_{xy}(x,y)$ denotes the probability that the chosen point lies below and leftwards of point (x,y)

In this example, $F_{xy}(x,y) = xy$

Now visualize $F_{xy}(x+h, y) - F_{xy}(x, y)$

This is the probability of the point lying in the small strip below y and b/w x and $x+h$.

$$\lim_{h \rightarrow 0} \frac{F_{xy}(x+h, y) - F_{xy}(x, y)}{h} = \frac{\partial F_{xy}(x, y)}{\partial x}$$

$$\frac{\partial F_{xy}(x,y)}{\partial y} = \frac{F(x,y+h) - F(x,y)}{h}$$

$$f_{xy}(x,y) = \frac{\partial^2 F_{xy}(x,y)}{\partial x \partial y} \quad \text{represents the joint probability density fn}$$

$$F_{xy}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{xy}(s,t) ds dt$$

$$f_{xy}(x,y) = f_x(x) f_y(y)$$

$$F_{xy}(x,y) = F_x(x) F_y(y)$$

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x,y) dy$$

$$f_y(y) = \int_{-\infty}^{\infty} f_{xy}(x,y) dx$$

Towards $E[g(x,y)]$

$$E[g(x)] = \sum_x g(x) p_x(x)$$

$$p_z(z) = \sum_{\substack{x \\ g(x)=z}} p_x(x)$$

x

$$\text{Let } z = g(x,y)$$

$$p_z(z) = \sum_{\substack{x,y \\ g(x,y)=z}} p_{xy}(x,y) = \sum_{\substack{(x,y) \\ g(x,y)=z}} p_{xy}(x,y) = \sum_{x,y} p_{xy}(x,y)$$

$$Q) E[ax + by + c]$$

$$= aE[x] + bE[y] + c$$

$$\sum_{x,y} (ax + by + c) p_{xy}(x,y) = a \sum_{x,y} x p_{xy}(x,y) + b \sum_{x,y} y p_{xy}(x,y) + c \sum_{x,y} p_{xy}(x,y)$$

$$= aE[x] + bE[y] + c //$$

$$E[g(x,y)] :$$

$$\text{Let } z = g(x,y)$$

$$E[z] = \sum_z \sum_{\substack{x,y \\ g(x,y)=z}} z p_{xy}(x,y)$$

$$= \sum_{x,y} g(x,y) p_{xy}(x,y)$$

For cont. rand variables

$$E[g(x,y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{xy}(x,y) dx dy$$

Examples

- 1) Suppose X and Y are continuous independent random variables. Let $W = \max(X,Y)$ and $Z = \min(X,Y)$

$$F_W(w_1) = P(W \leq w_1) = P(X \leq w_1 \cap Y \leq w_1) = \int_{-\infty}^{w_1} \int_{-\infty}^{w_1} f_{xy}(w_1, w_1) dx dy$$

$$= F_X(w_1) F_Y(w_1)$$

$$f_w(w) = \frac{d[F_x(w_1)F_y(w_1)]}{dw_1} = f_x(w)F_y(w) + F_x(w)f_y(w)$$

$$\begin{aligned}\overline{F_2(z)} &= P(Z \geq z_1) \\ &= P(X \geq z_1 \cap Y \geq z_1) \\ &= \overline{F_{xy}(z_1, z_1)}\end{aligned}$$

$$= 1 - F_{xy}(z_1, z_1) = 1 - F_x(z_1)F_y(z_1)$$

Let $z = x + y$

$$P_2(z) = \sum_{x,y: x+y \leq z} P_x(x) P_y(y)$$

Covariance of X and Y

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

$$V_{xy}[X] = E[(X - E[X])^2]$$

$$g(x, y) = (x - E[X])(y - E[Y])$$

$$E[(X - E[X])(Y - E[Y])] = \sum_{x,y} g(x, y) P_{xy}(x, y)$$

When Covariance is 0, the rand vars are said to be uncorrelated

If X & Y are independent $\rightarrow$ they are uncorrelated